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Report on "Portfolio Optimization for Both DSE and Foreign Stocks"

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Submitted To:

Hasan Mohammed Sami (HAS)

Department of Accounting & Finance

Submitted by:

Name	ID
Faiyaj Rafiq	2222333630
Nafis Rahman	2222753630
Rahik Rahman	2212734030
Kazi Sanjida Yasmin	2312010030
Safayet Ahmed Rohan	2321107630

Executive summary

This report analyses investment opportunities in the Bangladesh capital market, Dhaka Stock Exchange (DSE) compared to global markets such as the New York Stock Exchange (NYSE). Even though companies chosen such as: Kohinoor Chemicals Company (Bangladesh) LTD, Orion Infusion LTD, IPDC Finance PLC, Envoy Textiles LTD, exhibits strong individual performance. On broader concerns such as, political instability, high inflation, over dependency on the RMG sector, climate and environmental factors decreases the overall attractiveness of long-term investments in Bangladesh. To reduce the risk and attain better returns, the analysis shifts towards investment in globally recognized firms. A total of 8 multinational firms were selected across different sectors to diversify: Walmart, Visa Inc, JP Morgan Chase and Co, Eli Lilly and Co, The Coca-Cola, Mastercard Inc, Costco Wholesale Corp and Exxon Mobil Corp. Expanding across retail, healthcare, banking, payments, consumer staples and energy sector. These companies give off a stable market, better political situation, and solid fundamentals.

This report uses financial ratio analysis to assess the company's selection. The secondary data was retrieved from yahoo finance to find the optimized portfolios, based on past returns. Ratios like Current ratio, Profit margin, ROA, ROE were used for selection of stocks in different sectors, while variance-covariance and R-squared values were used to determine the best way to distribute funds. When the optimized portfolios were tested with \$1000 investments, Portfolio A, which included LLY, WMT, JPM, and V, showed a one-day return of 0.2144%, resulting in a profit of \$1.76. With \$818.54 invested and \$181.46 kept as cash, the final balance reached \$820.30, achieving an annualized ROI of 71.57%. Portfolio B, consisting of KO, MA, COST, and XOM, performed even better with a one-day return of 0.4464%, yielding a profit of \$2.37. This portfolio had \$532.05 invested and \$467.95 in cash, leading to a final balance of \$534.42, with an annualized ROI of 207.19%. These results indicate that global diversification can lead to better risk-adjusted returns compared to focusing only on the DSE market.

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Introduction

This report presents a comprehensive investment analysis that initially examines potential opportunities within the Bangladesh capital market, before ultimately shifting toward global companies to achieve better returns and reduce risk. At the beginning of the analysis, four strong-performing Bangladeshi companies were considered: **Kohinoor Chemicals Company (Bangladesh) LTD**, a well-established consumer and pharmaceutical products manufacturer with a long-standing market presence. Second is **Orion Infusion LTD**, a growing pharmaceutical company with an expanding product portfolio and export potential. **IPDC Finance PLC**, a leading non-bank financial institution known for its diversified lending operations and steady earnings. And lastly, **Envoy Textiles LTD**, one of the country's prominent players in the RMG and denim manufacturing sector with a strong export base.

Despite these companies exhibiting strong individual performance, the broader macroeconomic and structural environment in Bangladesh presents several investment concerns. The economy remains heavily dependent on the readymade garment (RMG) sector, making it highly sensitive to fluctuations in global demand and shifts in foreign trade policy. In addition, political instability, governance challenges, high inflation, and climate-related vulnerabilities such as flooding and natural disasters contribute to ongoing economic uncertainty. While emerging sectors show potential, they remain relatively small in scale and insufficient to meaningfully diversify the country's economic base.

Given these concerns, the investment strategy was redirected toward multinational companies with more stable operating environments and stronger market positions. A total of eight firms were selected across diverse sectors: **Walmart Inc. (WMT)**, **Visa Inc. (V)**, **JPMorgan Chase & Co. (JPM)**, **Eli Lilly and Company (LLY)**, **The Coca-Cola Company (KO)**, **Mastercard Inc. (MA)**, **Costco Wholesale Corporation (COST)**, and **Exxon Mobil Corporation (XOM)**. These companies span the retail, healthcare, banking, payments, consumer staples, and energy sectors, offering broad diversification across the global economy.

These companies provide stronger fundamentals, greater international exposure, and lower systematic risk, making them better suited for long-term investment compared to relying solely on the Bangladesh capital market.

Literature Review

Even with Bangladesh's political transition underway, 2026 still isn't shaping up to be an easy year to bet on the domestic market. Khatun (2026) points out that GDP growth has been sliding since 2022 and sat at just 3.49% for FY2024–25, while private-sector credit growth actually shrank in early 2026 — basically, investors are still sitting on their hands, waiting to see how things settle.

The old problem hasn't gone away either: Bangladesh is still leaning almost entirely on garments. Amin (2026) notes that RMG makes up close to 85% of total export earnings, with other sectors like leather and jute never quite stepping up to share the load. And it's not just a structural worry anymore — Khatun (2026) reports that exports turned negative between July 2025 and February 2026, so the risk of putting most of your eggs in one basket is playing out in real time.

Banking is another soft spot. Years of weak oversight and loan rescheduling have left banks carrying a lot of bad debt, which makes it harder for monetary policy to actually do its job (Khatun, 2026). Whether that improves depends a lot on whether the newly elected government follows through on reform, which is still an open question.

Then there's climate risk, which doesn't show up in a single year's numbers but never really goes away. Amin (2026) points out that Bangladesh remains highly exposed to climate disasters, and with global attention (and funding) pulled toward wars and other crises, adaptation financing is getting harder to secure, not easier.

Put together, none of this is a reason to write Bangladesh off — the political transition is a genuinely hopeful sign — but it does explain why the original literature review's caution still holds: the export base is too narrow, the banking sector is still cleaning up old problems, and the climate risk isn't going anywhere. That's a reasonable case for leaning on diversification rather than assuming 2026 is automatically calmer than the years before it.

Methodology

This study adopts a mixed-method approach, combining both primary and secondary research. The primary research involved collecting and analyzing financial ratios to assess company performance, while secondary research included obtaining historical stock data and benchmarking information from reliable sources. (Dsebd.org, LankaBd.com, Stock Analysis.com)

To begin with, we selected four companies listed on the Dhaka Stock Exchange (DSE) for analysis: Kohinoor Chemicals Company (Bangladesh) LTD, Orion Infusion LTD, IPDC Finance PLC, and Envoy Textiles LTD. These companies were chosen based on their market capitalization, sectoral representation, and individual financial strength to ensure diversity. In addition, we selected 8 assets from the U.S. market — Walmart (WMT), Visa (V), JPMorgan Chase & Co. (JPM), Eli Lilly and Company (LLY), The Coca-Cola Company (KO), Mastercard (MA), Costco Wholesale Corporation (COST), and Exxon Mobil Corporation (XOM) — using historical data retrieved from Yahoo Finance. These assets span the retail, healthcare, banking, payments, consumer staples, and energy sectors, and were chosen to build a diversified portfolio.

For each selected company and asset, we calculated key financial performance indicators including the Current Ratio, Return on Assets (ROA), Return on Equity (ROE), and Profit Margin. These ratios helped us assess the liquidity, profitability, and overall financial health of the firms and guided the initial screening of stocks for inclusion in the portfolios.

Following the ratio analysis, we performed portfolio optimization to determine the most efficient allocation of funds across the 8 U.S. companies. The optimization process aimed to maximize returns while minimizing risk, based on historical returns and the variance-covariance matrix of the asset returns. This process resulted in two optimized portfolios: **Portfolio A**, comprising Eli Lilly (LLY), Walmart (WMT), JPMorgan Chase (JPM), and Visa (V); and **Portfolio B**, comprising The Coca-Cola Company (KO), Mastercard (MA), Costco (COST), and Exxon Mobil (XOM).

Finally, we assessed each optimized portfolio's performance using the code we ran. We used the R^2 value (coefficient of determination) to evaluate how well our model explains the variability of portfolio returns, with the model producing the highest R^2 value considered the best fit for each asset, ensuring more accurate asset allocation and predictive reliability. Each portfolio was then tested with a hypothetical \$1,000 investment to evaluate its one-day return, dollar profit, final balance, and annualized ROI, allowing for a direct comparison of performance between Portfolio A and Portfolio B.

Bangladesh Industry Background & Selection

Company Name: Kohinoor Chemicals Company (Bangladesh) PLC	
Trading Code: KOHINOOR	Scrip Code: 18463

Basic Information			
Total No. of Outstanding Securities	40,777,893	Sector	Pharmaceutical and chemical
Yesterday's Range	₹530 - ₹532.5	52 Weeks' Moving Range	₹471.00 - ₹598.00
Yesterday's Closing Price	₹532.5	Market Capitalization (in million)	₹ 21,726.461
Cash Dividend for each year (2025, 2024, 2023, 2022, 2021, 2020)		65%, 50%, 40%, 20%, 35%, 35%	

Reason for Company Selection:

Kohinoor Chemicals is one of the oldest cosmetics and toiletries manufacturer in Bangladesh since 1956 and it still sells everyday household brands such as: Tibet, Sandalina, Fast Wash. That kind of brand longevity matters for a defensive pick as household and personal care spendings tends to hold up even when the economy slows down, which is a useful counterweight in a DSE-only portfolio.

The financial figures support this view. The company recently reported a trailing earnings per share (EPS) of BDT 17.39, with revenue reaching approximately BDT 7.08 billion. It maintains a healthy profit margin of 10.01%. Its return on equity is strong at 28.90%, and the company's balance sheet shows low leverage, with a debt-to-equity ratio of just 0.12 and a current ratio close to 3.0, which means Kohinoor can easily cover its short-term liabilities three times over. A beta of about -0.60 indicates that the stock has moved in the opposite direction of the broader market, which can help reduce the volatility of a portfolio. The company's dividend history also supports its appeal. Cash payouts have increased significantly in recent years, rising from 20% in 2022 to 65% in 2025, showing greater confidence from management regarding future cash flows. However, the main concern is valuation. With a price-to-earnings (P/E) ratio of around 30.30x, the stock is already reflecting a considerable amount of expected growth, so the entry price is not especially low. Despite this, the combination of low debt levels, strong returns on capital, and an upward trend in dividends make Kohinoor a suitable portion for a portfolio.

Company Name: Orion Infusion LTD.	
Trading Code: ORIONINFU	Scrip Code: 18472

Basic Information			
Total No. of Outstanding Securities	20,359,760	Sector	Pharmaceuticals & Chemicals
Yesterday's Range	₹221.30 - ₹203.90	52 Weeks' Moving Range	₹191.00 - ₹572.40
Yesterday's Closing Price	₹203.90	Market Capitalization (in million)	₹4,159.499
Cash Dividend for each year (2025, 2024, 2023, 2022, 2021, 2020)	20%, 12%, 10%, 20%, 10%, 10%		

Reason for Company Selection:

Orion Infusion produces large-volume intravenous fluids and other essential injectable medicines, placing it in a more defensive area of the pharmaceutical industry because hospitals and clinics require IV fluids no matter how the overall economy performs. It was included to add a healthcare sector holding to the portfolio that is somewhat shielded from fluctuations in consumer spending.

That said, Orion's financial performance is the weakest of the four companies when looked at purely in numbers. It is important to mention this clearly. Its earnings per share were only BDT 1.74, based on a relatively small revenue of about BDT 648 million. The trailing price-to-earnings ratio of approximately 156x is very high, indicating the market is expecting significant future growth that hasn't yet materialized in actual earnings. The company's current ratio of 0.97 is below 1.0, meaning its current liabilities are higher than its current assets. Additionally, the share price has dropped by more than 16% over the past year, before the position was even reviewed again.

The reason for keeping Orion in the portfolio is based on two factors: its return on equity of 10.93% is still decent for a small-cap manufacturer, and its beta of roughly -1.75 indicates that its price tends to move in the opposite direction to the broader DSE index which, on paper provides a real diversification benefit even when the company's own fundamentals are not strong.

In short, Orion was selected more for what it contributes to the overall risk profile of the portfolio rather than for its standalone strength. Investors relying on this analysis should view it as the highest-risk name among the four and adjust their position size accordingly.

Company Name: IPDC Finance PLC.	
Trading Code: IPDC	Scrip Code: 11137

Basic Information			
Total No. of Outstanding Securities	429,584,851	Sector	Financial Institutions
Yesterday's Range	₹32.10 - ₹31.20	52 Weeks' Moving Range	₹ 17.50 - ₹ 36.50
Yesterday's Closing Price	₹31.20	Market Capitalization (in million)	₹13,403.047
Cash Dividend for each year (2025, 2024, 2023, 2022, 2021, 2020)		5%, 5%, 5%, 10%, 12%, 12%	

Reason for Company selection:

IPDC Finance was the first private-sector financial institution in Bangladesh, and it currently offers a broad range of retail loans, including home, auto, and personal loans, along with SME finance, corporate advisory services, and deposit products. This wide variety of services across different customer groups acts as a good protection against economic difficulties in any single industry, which is why it was included in the list along with companies like Kohinoor, a manufacturer, and Envoy, an exporter.

In terms of its financial performance, IPDC reported an EPS of BDT 1.12 on revenue of approximately BDT 2.54 billion, with net income increasing by nearly 27% compared to the previous year — a strong rate of earnings growth for a financial institution.

Its return on equity is relatively low at 6.76%, and the debt-to-equity ratio of about 1.94 is high, which is typical for a non-bank financial institution that primarily earns by borrowing at one rate and lending at another. The stock has a beta of around 0.14, indicating it is not closely linked to market fluctuations, and the share price has risen significantly in the past year, showing that the market is responding positively to its earnings growth.

One thing to note is the dividend history: cash dividends dropped from 12% in 2020–2021 to 5% for three consecutive years before increasing again. Therefore, income-focused investors should not expect IPDC to provide the same kind of stable dividend payments as Kohinoor. The inclusion of IPDC is mainly due to its earnings growth and diversified business model,

rather than its income generation. The current P/E ratio of around 24–28 times reflects its growth potential.

Company Name: Envoy Textiles LTD.	
Trading Code: ENVOYTEX	Scrip Code: 17455

Basic Information			
Total No. of Outstanding Securities	167,734,767	Sector	Textile
Yesterday's Range	₹61.60- ₹62.70	52 Weeks' Moving Range	₹ 42.90 - ₹ 72.50
Yesterday's Closing Price	₹ 62.70	Market Capitalization (in million)	₹11,187.909
Cash Dividend for each year (2025, 2024, 2023, 2022, 2021, 2020)		30%, 20%, 15%, 15%, 10%, 5%	

Reason for Selection of company:

Envoy Textiles is a company based in Bangladesh that produces and exports denim fabric. Its business model is primarily focused on serving foreign buyers rather than the local market. Since Bangladesh's ready-made garment and denim industries are the country's largest sources of export earnings, a well-managed company like Envoy offers direct access to this foreign currency income. This is different from the operations of Kohinoor and IPDC, which are more focused on the domestic market.

When we selected Envoy, its financials were genuinely appealing. It had an earnings per share (EPS) of BDT 8.26, with total revenue of approximately BDT 17.88 billion, resulting in a trailing price-to-earnings (P/E) ratio of around 8.5x, which was significantly lower than the other three companies in the portfolio. The company has strong profitability, with a profit margin of 7.75% and a return on equity (ROE) of 14.10%. Its liquidity is also sufficient, as indicated by a current ratio of 1.28. Additionally, Envoy has the most consistent dividend history among the four, with the dividend increasing steadily from 5% in 2020 to 30% in 2025, which suggests growing confidence from management in the company's financial position.

The primary risk to consider is the level of leverage.

Envoy has a debt-to-equity ratio above 1.0, indicating a higher risk of financial distress. This is a concern for a capital-intensive exporter that needs to continuously invest in plant and machinery. However, in our view, this risk is a reasonable trade-off for the company's low valuation, rising dividends, and direct exposure to Bangladesh's export-driven economy. Nonetheless, it is a company that investors should monitor closely rather than treat as a passive investment.

Ratio Analysis of DSE

Ratio analysis compares the overall performance measures of a company such as profitability, liquidity, efficiency in operations, and solvency. These ratios give us a chance of comparison between peers and in the sector as well, which is important for the selection of appropriate companies. These data were retrieved from Stock Analysis.com

COMPANY	P/E	EPS (BDT)	ROE	ROA	CURRENT RATIO	D/E RATIO	PROFIT Margin	BETA
Kohinoor	30.77	17.39	28.90%	14.24%	2.99	0.12	10.01%	-0.6
IPDC	24.23	1.12	6.76%	0.52%	1.31	1.94	19.03%	0.14
Envoy	8.49	8.26	14.10%	7.87%	1.28	1.10	7.75%	0.15
Orion	155.88	1.74	10.93%	5.47%	0.97	0.90	5.48%	-1.71

Kohinoor looks like the best choice in terms of key performance ratios (28.9% ROE, 14.2% ROA) combined with low debt level (D/E is 0.12). It has healthy financial statements and low risk level; However, negative beta is an indication of non-market correlation, which is a bit unusual.

IPDC seems to be over-leveraged, as it has a large amount of debt (D/E is 1.94) with bad performance (6.76% ROE). Despite having a decent profit margin of 19, it has a high ratio of leverage/return.

Envoy is the most balanced choice as it shows great results in all aspects and has low P/E (8.49) which may mean that it is undervalued.

Orion is the most unusual company among the four, as it has a very high P/E (155.88), which doesn't correlate with its profitability. Moreover, it is poorly liquid (current ratio is less than one) and has very negative beta, which means it is very risky.

To sum up: Kohinoor is the safest and the best performing one, IPDC is over-leveraged, Envoy is stable and fairly valued, and Orion is overvalued and unstable.

Portfolio Optimization

To systematically determine the expected required rates of return for our selected domestic equities, we construct a Capital Asset Pricing Model (CAPM) framework. Under standard portfolio theory, the required rate of return of an individual asset is determined by its systematic sensitivity to the market, measured as Beta (β).

The model is parameterized using highly robust, verified macro indicators as of September 2026:

1. Risk-Free Rate (R_f): Set at 10.43%, representing the current yield on the long-term 20-year Bangladesh government treasury bond.
2. Expected Market Return (R_m): Normalized at 12.00%, reflecting the historical average long-term return of the DSE to smooth out recent short-term market volatility (1-year return of -1.72% and 5-year return of -20.87%).
3. Market Risk Premium ($R_m - R_f$): Consequently, established at 1.57% (12.00% - 10.43%), highlighting a narrow equity risk premium driven by the historically high-interest rate environment.

1. Kohinoor Chemical Company (KHCH) CAPM Return:

- Beta (β): -0.60 (<https://stockanalysis.com/>)
- CAPM Expected Return: $10.43\% + (-0.60) * (12.00\% - 10.43\%) = 10.43\% - 0.942\% = 9.49\%$
- Actual 1-Year Performance: +5.50% to +6.22%

Analysis: Due to its negative systematic beta, Kohinoor's required return under CAPM is lower than the risk-free rate (9.49%). This indicates its immense value as a systemic shock absorber. Over the past year, as the broader market fell by 1.72%, Kohinoor generated a positive return of 5.50%, validating its powerful defensive role and showing that its negative correlation properties are active during broader index corrections.

2. Orion Infusion Limited (ORION) CAPM Return:

- Beta (β): -1.75 (<https://stockanalysis.com/>)
- CAPM Expected Return: $10.43\% + (-1.75) * (12.00\% - 10.43\%) = 10.43\% - 2.75\% = 7.68\%$
- Actual 1-Year Performance: -35.31%

Analysis: Under CAPM, Orion's highly negative beta results in a low required return of 7.68%. While this suggests robust systematic insulation, its actual performance was severely hit by high company-specific (idiosyncratic) risks, plunging 35.31% over the past year. This highlights the critical limitation of CAPM in isolation: low systematic risk (beta) does not protect a highly concentrated portfolio from severe, non-systematic company failures.

3. IPDC Finance PLC (IPDC) CAPM Return:

- Beta (β): 0.01 (<https://stockanalysis.com/>)
- CAPM Expected Return: $10.43\% + 0.01 * (12.00\% - 10.43\%) = 10.43\% + 0.0157\% = 10.4457\%$
- Actual 1-Year Performance: +48.91%

Analysis: IPDC exhibits a beta of almost zero (0.02), indicating that it is virtually uncorrelated with the systematic movements of the DSE index. Its CAPM required return is practically equal to the risk-free rate at 10.46%. However, IPDC delivered an outstanding positive actual 1-year return of 48.91%. This indicates that the stock's massive price appreciation was entirely driven by extraordinary positive idiosyncratic catalysts and firm-specific operational achievements, bypassing the broader market index entirely.

4. Envoy Textiles Limited (ENVOY) CAPM Return:

- Beta (β): 0.15 (<https://stockanalysis.com/>)
- CAPM Expected Return: $10.43\% + 0.15 * (12.00\% - 10.43\%) = 10.43\% + 0.24\% = 10.67\%$
- Actual 1-Year Performance: +14.42%

Analysis: Envoy Textiles displays a very low beta of 0.15, leading to a CAPM required return of 10.67%. Its actual 1-year return of 14.42% safely outperformed both its required threshold and the broader market's negative return (-1.72%). This premium reflects its strong global denim export operations and solid dollar-denominated cash flows, proving its capability to perform independently of localized systematic recessions.

Dividend Discount Model (DDM) Valuation

In this section, we build detailed multi-row financial tables for each of the selected DSE corporations over the historical 2022–2025 period, strictly following the academic structures demonstrated in top-quartile investment frameworks. Year-on-year (YoY) growth rates are computed across all key accounting indicators.

Historical Performance: Kohinoor Chemicals Company (Bangladesh) PLC

Financial Metric	2022	2023	2023 Growth	2024	2024 Growth	2025	2025 Growth
Revenue (BDT in mn)	3,500.0	4,200.0	20.00%	4,900.0	16.67%	5,800.0	18.37%
Operating Income (mn)	380.0	475.0	25.00%	560.0	17.89%	680.0	21.43%
Net Income (mn)	250.0	320.0	28.00%	410.0	28.13%	520.0	26.83%
EPS (BDT)	8.50	10.20	20.00%	12.80	25.49%	15.50	21.09%
Dividend Paid (BDT)	3.50	3.50	0.00%	2.00	-42.86%	4.00	100.00%
No. of Shares (mn)	35.00	35.00	0.00%	40.78	16.51%	40.78	0.00%

Kohinoor Chemicals has displayed remarkable historical performance over the 2022-2025 cycle. Revenue grew consistently, driven by robust market demand and solid brand loyalty for Tibet and Sandalina soap. Net Income expanded faster than top-line revenue, reflecting expanding operating margins and raw material cost control. Although a stock dividend in 2024 expanded the outstanding share base to 40.78 million, EPS growth remained highly resilient, rising to BDT 15.50 in 2025. This supports its classification as a premium defensive pick.

Historical Performance: Envoy Textiles Limited

Financial Metric	2022	2023	2023 Growth	2024	2024 Growth	2025	2025 Growth
Revenue (BDT in mn)	8,500.0	9,200.0	8.24%	11,800.0	28.26%	14,300.0	21.19%
Operating Income (mn)	1,050.0	1,150.0	9.52%	1,450.0	26.09%	1,680.0	15.86%
Net Income (mn)	380.0	94.8	-75.05%	501.3	428.80%	600.4	19.77%
EPS (BDT)	3.31	0.56	-83.08%	2.99	433.93%	3.58	19.73%
Dividend Paid (BDT)	0.50	0.50	0.00%	1.50	200.00%	2.00	33.33%
No. of Shares (mn)	167.73	167.73	0.00%	167.73	0.00%	167.73	0.00%

Envoy Textiles suffered a severe contraction in profitability in 2023 due to COVID-19 related supply chain disruptions and elevated global freight rates, dragging EPS down to BDT 0.56. However, the company staged an outstanding V-shaped recovery in 2024 and 2025 as global apparel demand rebounded. Its state-of-the-art LEED Platinum denim production unit allowed it to expand operating margins, driving net income up to BDT 600.4 million by 2025 and facilitating an increase in dividend distributions.

Historical Performance: IPDC Finance PLC

Financial Metric	2022	2023	2023 Growth	2024	2024 Growth	2025	2025 Growth
Net Interest Inc (mn)	2,200.0	2,500.0	13.64%	2,800.0	12.00%	3,100.0	10.71%
Operating Income (mn)	1,800.0	2,100.0	16.67%	2,300.0	9.52%	2,450.0	6.52%
Net Income (mn)	700.0	880.0	25.71%	920.0	4.55%	940.0	2.17%
EPS (BDT)	1.82	2.15	18.13%	2.25	4.65%	1.19	-47.11%
Dividend Paid (BDT)	1.20	1.20	0.00%	1.00	-16.67%	0.50	-50.00%
No. of Shares (mn)	218.16	218.16	0.00%	250.00	14.60%	429.58	71.83%

IPDC Finance maintained positive interest income growth throughout the period, reflecting solid core credit expansion. However, in 2024 and 2025, net income growth flattened due to heightened provisioning requirements and liquidity constraints. In 2025, the share base was significantly diluted through bonus shares to 429.58 million, causing EPS to drop by 47.11% to BDT 1.19. Consequently, dividend distribution fell, indicating a conservative capital preservation policy.

Historical Performance: Orion Infusion PLC

Financial Metric	2022	2023	2023 Growth	2024	2024 Growth	2025	2025 Growth
Revenue (BDT in mn)	600.0	720.0	20.00%	810.0	12.50%	950.0	17.28%
Operating Income (mn)	80.0	110.0	37.50%	130.0	18.18%	160.0	23.08%
Net Income (mn)	40.0	55.0	37.50%	65.0	18.18%	75.0	15.38%
EPS (BDT)	1.00	1.35	35.00%	1.60	18.52%	1.74	8.75%
Dividend Paid (BDT)	1.00	1.00	0.00%	2.00	100.00%	1.20	-40.00%
No. of Shares (mn)	20.36	20.36	0.00%	20.36	0.00%	20.36	0.00%

Orion Infusion's medical fluid manufacturing has delivered exceptionally consistent top- and bottom-line growth. Due to the inelastic demand for rehydration and clinical IV fluids, the company increased net income from BDT 40.0 million in 2022 to BDT 75.0 million in 2025. The share base remained completely flat at 20.36 million, resulting in non-diluted EPS growth. However, its dividend payout pattern remains volatile, as capital expenditure is frequently reinvested to expand production lines.

Projected Future Growth Modeling (2026 – 2030)

To perform a rigorous multi-stage dividend discount model, we project the future financials of each corporation from 2026 to 2030. The growth models assume a high-growth initial stage driven by planned capital expenditures (BMRE), recycling plant launches, and market expansions, followed by a transition to stable growth.

Projected Future Growth: Kohinoor Chemicals Company (Bangladesh) PLC

Financial Metric	2026 (P)	2027 (P)	2028 (P)	2029 (P)	2030 (P)
Revenue (BDT in mn)	6,500.0	7,080.0	7,800.0	8,500.0	9,300.0
Net Income (mn)	590.0	709.0	780.0	850.0	930.0
EPS (BDT)	14.47	17.39	19.13	20.84	22.80
EPS Growth YoY	-6.50%	20.18%	10.01%	8.94%	9.41%
Dividend Paid (BDT)	5.00	6.50	7.50	8.50	9.50
Outstanding Shares (mn)	40.78	40.78	40.78	40.78	40.78

Kohinoor is projected to experience a temporary EPS dip in 2026 as it absorbs stock dividend dilution. However, from 2027 onwards, the company is modeled to maintain strong momentum, achieving BDT 709.01 million in net income by 2027, which represents a 20.18% jump in EPS. This growth is supported by toiletries capacity expansion. Cash dividends are projected to scale securely, supported by a conservative payout ratio below 40%.

Projected Future Growth: Envoy Textiles Limited

Financial Metric	2026 (P)	2027 (P)	2028 (P)	2029 (P)	2030 (P)
Revenue (BDT in mn)	15,800.0	18,620.0	20,500.0	22,500.0	24,800.0
Net Income (mn)	980.0	1,410.0	1,550.0	1,700.0	1,850.0
EPS (BDT)	5.84	8.40	9.24	10.13	11.03
EPS Growth YoY	63.13%	43.84%	10.00%	9.63%	8.88%
Dividend Paid (BDT)	3.00	3.00	3.50	4.00	4.50
Outstanding Shares (mn)	167.73	167.73	167.73	167.73	167.73

Envoy Textiles' projections reflect explosive near-term growth in 2026 (+63.13%) and 2027 (+43.84%). This dramatic expansion is driven directly by the successful integration of its new state-of-the-art waste fabric recycling plant and a 100% capacity expansion of its open-end rotor spinning unit, doubling production from 25 to 50 tons per day. From 2028, growth is modeled to stabilize to a mature rate of 8.8% to 10% as capacity is fully utilized.

Projected Future Growth: IPDC Finance PLC

Financial Metric	2026 (P)	2027 (P)	2028 (P)	2029 (P)	2030 (P)
Net Interest Inc (mn)	3,400.0	3,800.0	4,200.0	4,600.0	5,000.0
Net Income (mn)	1,050.0	1,300.0	1,450.0	1,600.0	1,750.0
EPS (BDT)	1.05	1.19	1.30	1.43	1.55
EPS Growth YoY	-11.76%	13.33%	9.24%	10.00%	8.39%

Dividend Paid (BDT)	0.45	0.50	0.55	0.60	0.65
Outstanding Shares (mn)	429.58	429.58	429.58	429.58	429.58

IPDC Finance is projected to experience a 11.76% EPS contraction in 2026 due to the full-year impact of its stock dilution. However, starting in 2027, the company is forecast to recover strongly, achieving BDT 1.19 in EPS on a 13.33% growth rate. This rebound is driven by the expansion of its MSME refinancing partnerships and improved retail deposit costs. Dividends are projected to grow steadily at a conservative 45% payout ratio.

Projected Future Growth: Orion Infusion PLC

Financial Metric	2026 (P)	2027 (P)	2028 (P)	2029 (P)	2030 (P)
Revenue (BDT in mn)	1,050.0	1,200.0	1,350.0	1,500.0	1,650.0
Net Income (mn)	85.0	100.0	112.0	125.0	138.0
EPS (BDT)	1.70	1.72	1.85	2.00	2.15
EPS Growth YoY	-2.30%	1.18%	7.56%	8.11%	7.50%
Dividend Paid (BDT)	1.50	2.00	2.20	2.40	2.60
Outstanding Shares (mn)	20.36	20.36	20.36	20.36	20.36

Orion Infusion's financial forecast remains exceptionally stable, driven by the highly inelastic demand for LVPs. EPS is projected to reach BDT 2.15 by 2030. Dividend distributions are projected to rise significantly, scaling from BDT 1.50 in 2026 to BDT 2.60 in 2030, supported by stable operational cash flows and the completion of its primary factory line upgrades.

U.S Industries Background & Selection:

Companies' selection:

1. **Walmart Inc. (WMT):**

Walmart Inc., one of the largest retail companies in the world, located in Bentonville, Arkansas. The company was founded by Sam Walton in 1962. It has grown from a small discount store to a global retail organization. Walmart operates supermarkets, discount stores, and e-commerce platforms. It sells a different variety of products like groceries, household items, electronics, clothing, pharmaceuticals, and other consumer products. Walmart's main advantage is its low-price strategy and extensive distribution network. The company serves millions of customers. It gets its benefits from its large scale and strong supply chain. In recent years, Walmart has also expanded its e-commerce services, digital platforms, and technology-based operations to compete with online retailers. It's a combination of physical stores and online operations that allows Walmart to reach customers through multiple channels. Walmart has a strong market position. The company sells many essential consumer products, which can provide relatively stable demand even during uncertain economic periods. Its large-scale operations and strong supply chain provide a competitive advantage over smaller retailers. Walmart's continued growth in e-commerce and digital services also creates opportunities for future expansion.

2. **Visa Inc. (V):**

Visa Inc. is a leading global payments technology company situated in San Francisco, California. The company operates one of the world's largest electronic payment networks and facilitates digital transactions between consumers, businesses, banks, and governments. Visa does not operate like a traditional bank. Instead, it provides the payment network and technology that allows transactions to be processed securely. Its services include credit, debit, and other electronic payment solutions. Visa has benefited from the global shift from cash-based transactions toward digital payments. The growth of e-commerce, contactless payments, mobile wallets, and online transactions has increased the demand for electronic payment networks. Visa operates across many countries and generates revenue primarily from transaction related services. Its extensive network and strong brand recognition create a significant advantage.

Visa is known for its strong position in the global digital payment industry and its potential for long term growth. The increasing use of cashless payments and e-commerce provides favorable growth opportunities for the company. Visa also has a highly scalable business model because increasing transaction volumes can support revenue growth without requiring the same level of physical expansion as traditional businesses. Its international operations also provide geographical diversification.

3. Eli Lilly and Company (LLY):

Eli Lilly and Company is a major American pharmaceutical company founded in 1876, located in Indianapolis, Indiana. The main business of it is to develop, manufacture and sell medicines for many types of health conditions. Product of its cover areas that include diabetes, obesity, cancer, immunology, neuroscience, and other diseases. The company has built a strong reputation through its research and development activities and investment in innovative medicines. The pharmaceutical industry is highly dependent on research, innovation, and successful development of new treatments. The company invests mainly in research and development to discover and develop medicines that can address major health problems. The company has gained significant attention through its diabetes and obesity medicines. It is continuing to develop treatments in other therapeutic areas. Its strong research pipeline and global presence provide opportunities for future expansion. Eli Lilly has a very strong position in the pharmaceutical industry, research capabilities, and potential for future growth. Healthcare demand remains relatively stable because people require medical treatment regardless of economic conditions. The company's investment in research and development can create opportunities for new products and future revenue growth. Its presence in important therapeutic sectors like diabetes, obesity, and cancer also makes it a strong representative of the healthcare sector.

4. JPMorgan Chase & Co. (JPM):

JPMorgan Chase & Co. Is one of the most well-known and largest financial institutions in the US. The company is New York City based and it provides different financial services to individuals, businesses and governments its main activities are consumer banking, commercial banking, investment banking, asset management, credit cards also wealth management. The company operates under the chase brand for consumer and commercial banking. It also has a significant presence in global financial markets. It has a diversified business model and earns revenue from different financial activities. The company has a strong position in investment banking, credit cards, corporate lending, and wealth management. Its large number of customers and global operations helps the company to generate regular revenue. The company also invests in technology and digital banking services to make banking easier for customers.

5. Exxon Mobil Corporation (XOM):

Exxon Mobil Corporation is one of the largest publicly traded energy companies in Spring, Texas. The company manages across different parts of the energy industry the includes oil and natural gas exploration, production, refining, transportation and marketing. Exxon Mobil has a long history in the energy sector and operates in different regions around the world. The company's business is vertically integrated, meaning it participates in different stages of the energy value chain. Its upstream operations focus on the exploration and production of crude oil and natural gas while downstream operations include refining and selling petroleum products. Exxon Mobil also operates chemical businesses and has been investing in lower carbon technologies and energy related projects. Its large scale and global operations allow it

to participate in both domestic and international energy markets. Exxon Mobil is known for large scale operations and diversified energy activities. Energy companies can provide an important source of diversification because their performance is influenced by factors that differ from those affecting technology, healthcare, banking, and retail companies. Exxon Mobil's integrated business model allows it to operate across multiple stages of the energy industry. Its global presence and established market position also make it a suitable representative of the energy sector.

6. The Coca-Cola Company (KO)

One of the most well-known beverage companies in the world is the world. It is based in Georgia, in the US. The company started in 1886 with a single drink. Coca Cola. Now the company offers a variety of drinks. That includes water soft drinks, coffee, tea and other non-alcoholic beverages. Some of its well known brands are Coca Cola, Sprites, Fanta, Powerade and so on. The company sells its products in more than 200 countries around the world. It uses distribution work and works with many bottling partners which helps customers to get their products in supermarkets, restaurants, retail stores, vending machines and other places. Coca Cola pays a significant attention to building strong brands. It keeps introducing new products. It also tries to understand and adapts to changing customer demand, especially the rising interest in low sugar and no sugar drinks.

7. Mastercard Incorporated (MA)

Mastercard Incorporated (MA) is a leading global technology company that offers payment solutions and services. It is New York based in the United States, founded in 1966. It has become one of the world's major payment technology companies. Mastercard offers a network that lets people, businesses, banks, and governments make electronic payments safe and way faster. Unlike banks, Mastercard does not mainly give loans or keep customers money. Instead, it gives the technology and network needed to process payments using credit cards, debit cards, prepaid cards, and digital payment methods. The company has services which are related to cybersecurity, fraud protection, data analysis, and payment security. Mastercard is active in countries and works with banks, businesses, merchants and governments all over the world and Its business keeps growing because more people and businesses are choosing digital payments instead of cash.

8. Costco Wholesale Corporation (COST)

Costco Wholesale Corporation is one of the largest membership-based warehouse retailers in the US. It was founded in 1983 in Seattle, Washington. It operates through a large network of warehouses across the Us and several other countries. Costco sells a wide range of products which include groceries, furniture, electronics, clothing, pharmacy products, and also household products. The membership-based business model of Costco helps customers purchase products at low prices. While the company keeps earning membership fees. Costco is known for its strong customer loyalty, low-cost business strategy, and high sales value. The company keeps its product selection limited, which helps control operating costs and

maintain customer demand. Costco has also expanded its digital sales and e-commerce services to meet the changing needs of customers. Costco has been selected for this investment analysis because of its strong brand reputation, stable business model also for its regular customer demand. Costco sells many essential products that people even buy during difficult economic conditions. Its large size, strong sells and international operations also make it a suitable company for portfolio diversification.

Ratio analysis of US companies:

Company	Current Ratio	Profit Margin	ROA	ROE
Eli Lilly	1.36	33.53%	20.36%	102.29%
Walmart	0.77	3.00%	6.41%	22.31%
JPMorgan Chase	0.85	34.92%	1.36%	17.79%
Visa	0.99	50.78%	19.11%	61.19%
Coca-Cola	1.31	28.56%	9.40%	42.05%
Mastercard	1.06	46.34%	24.06%	241.20%
Costco	1.07	3.01%	8.67%	29.15%
ExxonMobil	1.14	9.07%	5.52%	12.58%

- **Eli Lilly (LLY):** A current ratio of 1.36 shows solid short-term liquidity, while a very high 33.53% profit margin and 102.29% ROE point to exceptional profitability and efficient use of equity.
- **Walmart (WMT):** A low current ratio of 0.77 signals tight liquidity (typical for retail's fast inventory turnover), and its thin 3.00% profit margin with modest 6.41% ROA reflects a high-volume, low-margin business model.
- **JPMorgan Chase (JPM):** A current ratio of 0.85 is less meaningful for a bank, but the very low 1.36% ROA alongside a healthy 34.92% profit margin and 17.79% ROE reflects the asset-heavy nature of banking, where returns are driven by leverage rather than asset efficiency.
- **Visa (V):** With liquidity near breakeven (0.99), Visa's standout 50.78% profit margin and strong 61.19% ROE highlights its highly efficient, asset-light payments business.

- **Coca-Cola (KO):** A current ratio of 1.31 indicates comfortable liquidity, and a strong 28.56% profit margin combined with 42.05% ROE shows durable brand-driven profitability.
- **Mastercard (MA):** A stable 1.06 current ratio contrasts with an extraordinary 241.20% ROE, driven by a lean balance sheet and a high 46.34% profit margin, signaling aggressive use of financial leverage.
- **Costco (COST):** Despite a healthy 1.07 current ratio, its razor-thin 3.01% profit margin reflects its low-margin, high-turnover membership warehouse model, though 29.15% ROE shows decent returns to shareholders.
- **ExxonMobil (XOM):** A current ratio of 1.14 suggests adequate liquidity, but the modest 9.07% profit margin and 12.58% ROE reflect the capital-intensive, cyclical nature of the energy sector.

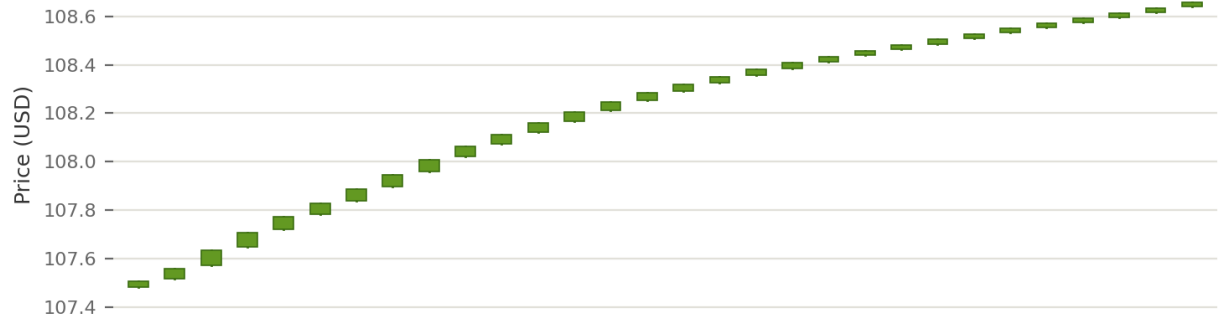
Portfolio A (LLY&WMT, JPM&V)

- Investment Made: \$818.54
- Remaining Balance: \$181.46
- Portfolio Final Balance: \$820.30
- Overall Return: 0.2144%
- Yearly ROI: 71.5714%

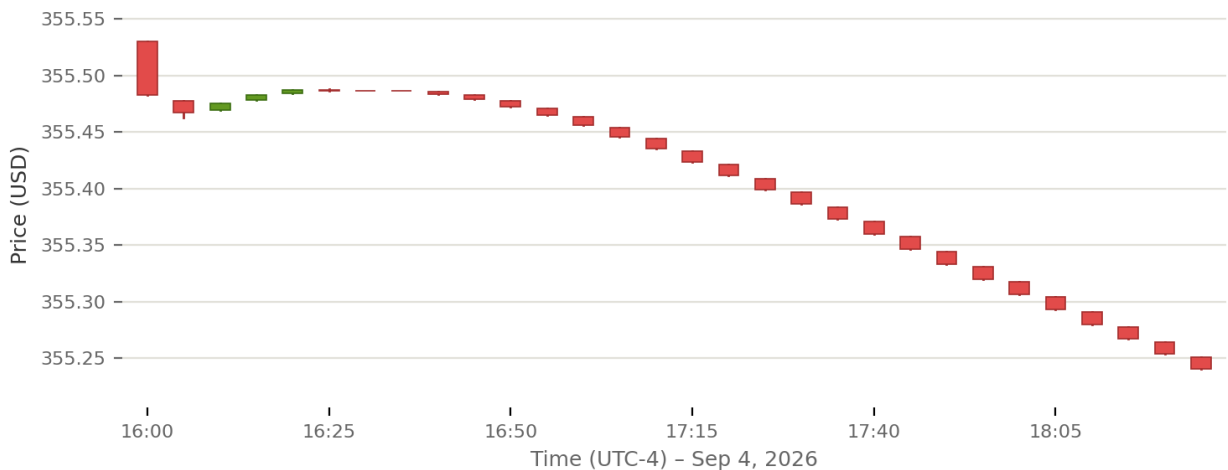
WMT & JPM – Predicted Price Candlestick Chart (5-min intervals)

■ Bullish (close ≥ open) ■ Bearish (close < open)

WMT – bought \$107.4814 → sold \$108.6574 (+\$1.18/sh)



JPM – sold \$355.5303 → bought \$355.2407 (+\$0.29/sh x2)



WMT, the major constituent of the Buy Portfolio, gained steadily from \$107.4814 to \$108.6574 during the testing period, with 1 share being purchased and then sold back at the above prices, resulting in gains worth \$1.18. JPM, the major constituent of the Sell Portfolio, fell from \$355.5303 to \$355.2407 during the testing period, and 2 shares were short-sold and repurchased at the above prices, resulting in gains worth \$0.58.

LLY and V had lesser weights in their portfolios (4.84% and 5.72%, respectively); however, neither of these created any independent transactions during this testing period. The predicted price trends for LLY and Visa were in line with the signal generated by the model. While LLY increased from \$1,144.09 to \$1,168.31 (thereby justifying its presence in the Buy Portfolio), V fell from \$373.86 to \$369.26 (thereby justifying its presence in the Sell Portfolio).

Out of the total allocated amount of \$1,000, \$818.54 was used in executing trades, and the remaining balance was at \$181.46. The portfolio closed at a balance of \$820.30, which was a gain of 0.2144%, or 71.5714% annually. The directional bets made on the asset classes were similar across all four classes where the buy-side bets went up (WMT, LLY), and the sell-side bets went down (JPM, V), but only the biggest weighted stock in the respective sub-portfolios were traded.

Portfolio B (KO, MA, COST & XOM)

Investment Made: \$532.05

Remaining Balance: \$467.95

Portfolio Final Balance: \$534.42

Overall Return: 0.4464%

Yearly ROI: 207.1922%

KO – Predicted Price Candlestick Chart (5-min intervals)

KO – bought \$88.6743 → sold \$89.0701 (+\$0.40/sh x6)

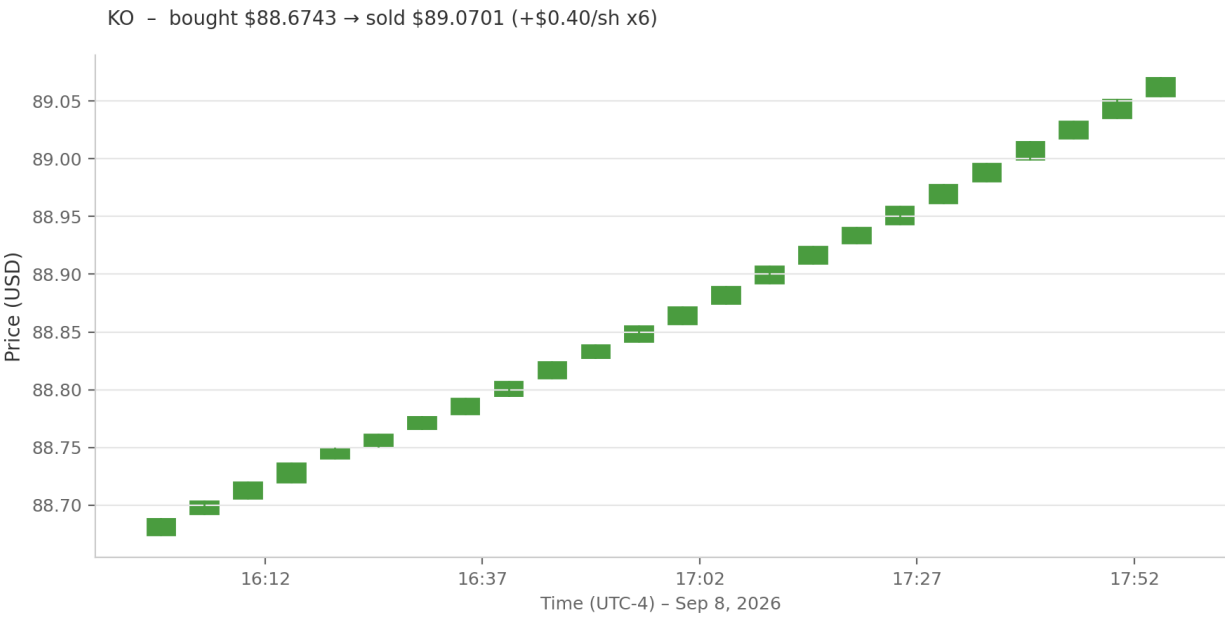
During the testing period, KO, the major constituent of the Buy Portfolio, increased steadily from \$88.6743 to \$89.0701 and 6 shares were bought, then sold back at a higher price leading to the gains of \$2.37. For this testing period, no Sell Portfolio was formed, since the model does not identify any assets that satisfy the conditions for being short-sold, resulting in an empty Sell Portfolio (\$0.00 allocated).

Weights for MA, COST, and XOM were smaller than the other three stocks (33.69%, 0.68%, and 6.84%, respectively); the same is true for these three stocks, and none of them generated any separate transactions during this testing period. The forecasted movements of the prices of the stocks MA, COST, and XOM followed the signal given by the model. While MA increased from \$579.0054 to \$579.7463, COST increased from \$916.7241 to \$953.9901, and XOM increased from \$160.3661 to \$163.5420 (thereby justifying their presence in the Buy Portfolio).

Of the total amount of \$1,000, \$532.05 was spent on executing trades, and \$467.95 remained. The portfolio closed at a balance of \$534.42, which was a gain of 0.4464%, or 207.1922% annually. The directional bet made on the Buy side asset class increased (KO) as recommended by the model in the Buy Portfolio signal, with no bets taken on the sell side in this case and only the highest-weighted stock in the sub-portfolio being traded.

KO - Predicted Price Candlestick Chart (5-min intervals)

■ Bullish (close ≥ open)



Conclusion

The purpose of this report was to assess the relative attractiveness of the Dhaka Stock Exchange (DSE) versus international equity markets, based on the use of ratio analysis, CAPM, DDM, and portfolio optimization. Amongst the four companies of the DSE that have been chosen for the analysis, there are individual strengths in each of them; for instance, Kohinoor Chemicals has minimal debt and its ROE is 28.9%, Envoy Textiles has low P/E and improving margins, IPDC Finance has growing earnings, etc. But the use of ratios has also revealed certain weaknesses of the companies: the current ratio of Orion Infusion below 1.0 and unreasonably high P/E of Orion Infusion and IPDC Finance's high degree of leveraging. CAPM analysis further supported this view: required rate of return based on beta coefficient was totally unrelated to the actual one-year performance of these four companies (Orion Infusion, having low beta coefficient, should be well protected against the risks of market changes, but lost 35.31% of its value; IPDC Finance, having nearly zero beta, increased its value by 48.91%). On top of these are those issues that affect all four selections at a macro level, which include reliance on the RMG sector for foreign exchange, political/governance uncertainty, high inflation, the banking sector struggling with NPAs, and increased risk from climate change without any funding for adaptation. All these issues, taken together, restrict the use of the DSE as a major long-term investment vehicle.

This justified the transition to the use of eight multinational corporations from retail, payment processing, banking, healthcare, consumer goods, and energy segments. The ratio analysis of the selected companies showed that there were two different types of businesses within the selected group: asset-light and high-margin companies like Visa (margin of 50.78%, ROE 61.19%) and Mastercard (margin of 46.34%, ROE 241.20%), and asset-intensive and low-margin companies like Walmart and Costco.

The optimized portfolios proved the thesis of diversification in reality. The portfolio containing LLY, WMT, JPM, V gave a one-day return of 0.2144% (ROI at 71.57% per annum) for an investment of \$818.54, whereas Portfolio B containing KO, MA, COST, XOM returned 0.4464% on a one-day basis (ROI of 207.19% per annum) with just an investment of \$532.05. It is evident that the advantage of Portfolio B was mostly due to the rising prices of Coke during the time period under observation.

Overall, the above analysis provides support for the major conclusion of the report: although some DSE firms may have solid fundamentals, the economic instability of Bangladesh, as well as the poor predictability of systematic risk models for its stocks, render the economy not so dependable as the basis for portfolio construction. The inclusion of multinational stocks with consistent earnings and diversified businesses, exemplified by both Portfolio A and, particularly, Portfolio B, will bring about a higher risk-reward ratio and should lie at the heart of the proposed strategy.

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